

BIO-MEDICAL PROJECT TOPICS IN AI, IOT, AND EMBEDDED SYSTEMS FOR HEALTHCARE & MONITORING

1. SMART WEARABLE FOR REAL-TIME RESPIRATORY AND SPEED TRACKING DURING PHYSICAL ACTIVITY
2. AI-INTEGRATED SMART PILLOW FOR COMPREHENSIVE HEALTH MONITORING AND SLEEP QUALITY ANALYSIS
3. NON-CONTACT AI-POWERED PATIENT MONITORING SYSTEM FOR INFECTIOUS DISEASE SURVEILLANCE
4. HEART RATE VARIABILITY AND AI-BASED DRIVER DROWSINESS DETECTION AND ALERT SYSTEM
5. AI-DRIVEN GAIT ANALYSIS USING WEARABLES FOR EARLY DETECTION OF NEUROLOGICAL DISORDERS
6. BIOMETRIC FINGERPRINT-BASED CENTRALIZED MEDICAL DATA ACCESS AND MANAGEMENT SYSTEM
7. VISUAL FOCUS-AWARE AI SYSTEM FOR ENHANCED HUMAN-ROBOT INTERACTION IN CARE FACILITIES
8. TELE-HEALTH ENABLED WEARABLE SENSOR NETWORK FOR REMOTE PATIENT MONITORING
9. SMART REHABILITATION SYSTEM FOR MUSCLE CRAMPS IN MILITARY AND ATHLETES USING EMBEDDED SENSORS
10. SWEAT COMPOSITION ANALYSIS USING AI FOR EARLY DETECTION OF MULTIPLE HEALTH CONDITIONS
11. NON-INVASIVE SMART SCANNER FOR EARLY DETECTION OF BONE HEALTH AND DISORDERS
12. BABY JAUNDICE LEVEL DETECTION USING OPTICAL SENSORS AND MACHINE LEARNING
13. RETINAL IMAGE-BASED PREDICTIVE SYSTEM FOR HEART ATTACK AND HYPERTENSION RISK ASSESSMENT

14. FOOT PRESSURE MAPPING SYSTEM FOR EARLY DIABETIC FOOT DIAGNOSIS AND MONITORING
15. PULSE TRANSIT TIME-BASED WEARABLE FOR CONTINUOUS CARDIOVASCULAR HEALTH MONITORING
16. AI-POWERED LOW-COST FOOT ULCER DETECTION SYSTEM FOR DIABETIC PATIENTS
17. NERVE SIGNAL MAPPING SYSTEM FOR PRE-DIAGNOSIS OF NEURAL DISORDERS USING ML
18. DEEP LEARNING-BASED PRE-STROKE PREDICTION SYSTEM USING VITAL SIGN TRENDS
19. HEART RATE-CONTROLLED AUDIO THERAPY SYSTEM FOR BLOOD PRESSURE REGULATION
20. AI-BASED PATIENT MONITORING AND SMART MEDICINE RECOMMENDATION SYSTEM
21. GAZE-CONTROLLED HOME AUTOMATION SYSTEM FOR PARALYZED INDIVIDUALS USING EMBEDDED VISION
22. IOT-BASED ADVANCED SOCIAL DISTANCING AND MASK DETECTION ALERT SYSTEM
23. MICROSCOPIC BLOOD IMAGE PROCESSING SYSTEM FOR EARLY CANCER CELL IDENTIFICATION
24. SMART HIP AND ABDOMEN SENSOR NETWORK FOR FALL DETECTION AND EMERGENCY RESPONSE
25. BREATHING PATTERN RECOGNITION SYSTEM FOR SLEEP QUALITY AND APNEA ANALYSIS
26. WEARABLE HIP SENSOR FOR ELDERLY MONITORING WITH LCD FEEDBACK SYSTEM
27. BRAIN WAVE ANALYSIS AND LEG MOVEMENT INTEGRATION FOR DISABLED COMMUNICATION AID
28. SMART MENTAL HEALTH MONITORING SYSTEM FOR PSYCHOLOGICAL DISORDER PREDICTION

29. REAL-TIME EMERGENCY RESPONSE COORDINATION SYSTEM IN SMART HOSPITALS
30. ICU INTEGRATED RESPIRATORY AND BODY TEMPERATURE MONITOR WITH ALARM SYSTEM
31. IOT-BASED REMOTE HEARTBEAT AND TEMPERATURE MONITORING FOR RURAL PATIENTS
32. AI-POWERED INFANT MONITORING TO REDUCE SUDDEN INFANT DEATH SYNDROME (SIDS) RISK
33. PERITONEAL DIALYSIS MONITORING AND ALERT SYSTEM FOR KIDNEY FAILURE PATIENTS
34. CENTRALIZED CARDIAC HEALTH MONITORING SYSTEM USING CLOUD AND ML
35. MYOELECTRIC SIGNAL PROCESSING SYSTEM FOR ARTIFICIAL HAND MOTOR CONTROL
36. COMPUTER-AIDED DESIGN AND FABRICATION OF ADAPTIVE LOWER LIMB PROSTHESIS
37. WIRELESS FM-BASED BIO-TELEMETRY SYSTEM FOR MOBILE PATIENT MONITORING
38. RETINAL IMAGE MOSAICKING SYSTEM FOR ADVANCED EYE DISEASE DIAGNOSIS
39. VOICE RECOGNITION-CONTROLLED SMART WHEELCHAIR FOR MOBILITY-IMPAIRED USERS
40. AI-BASED ERYTHROCYTE SHAPE ANALYSIS UNDER PATHOLOGICAL CONDITIONS
41. ALL-IN-ONE MASTER HEALTH CARD SYSTEM FOR LIFETIME PATIENT HISTORY ACCESS
42. TUMOR AND VESSEL DETECTION IN LIVER CT IMAGES USING IMAGE PROCESSING
43. PORTABLE BLOOD OXYGENATOR SYSTEM WITH SMART MONITORING FEATURES

44. ELECTRO-MECHANICAL FOOT TO KNEE MASSAGING SYSTEM FOR DIABETIC NEUROPATHY
45. SKIN TONE MONITORING SYSTEM FOR ANEMIA AND DEHYDRATION ANALYSIS
46. AI-POWERED ODOR RECOGNITION SYSTEM FOR DETECTING CHEMICAL EXPOSURE SYMPTOMS
47. DSP-BASED DIGITAL HEARING AID SYSTEM FOR AGE-RELATED HEARING LOSS
48. WEARABLE PROSTHETIC GUIDANCE SYSTEM FOR BLIND AMPUTEES WITH GPS NAVIGATION
49. SMART REAL-TIME HEALTH TRACKING PILLOW WITH INTEGRATED VITAL SENSORS
50. MACHINE LEARNING-DRIVEN SWEAT ANALYZER FOR MULTI-DISEASE DIAGNOSIS
51. IOT AND AI-BASED PLATFORM FOR FOOT INFECTION AND DISEASE CLASSIFICATION
52. AUTOMATED ECG SIGNAL ANALYZER FOR ARRHYTHMIA DETECTION USING DEEP LEARNING
53. DEEP LEARNING-BASED AUTISM DETECTION USING FACIAL AND VOCAL BIOMARKERS
54. CONTINUOUS GLUCOSE MONITORING SYSTEM USING WEARABLE NON-INVASIVE SENSORS
55. GABA LEVEL MONITORING SYSTEM USING AI FOR NEUROLOGICAL HEALTH PREDICTION
56. AI AND IOT-BASED COMPREHENSIVE MULTI-DISEASE PREDICTION USING BODY SENSOR NETWORK
57. SMART VOICE-ACTIVATED PATIENT BED FOR MOVEMENT ASSISTANCE AND VITAL MONITORING
58. AI-BASED WOUND HEALING PREDICTION SYSTEM USING U-NET IMAGE SEGMENTATION

59. SMART INSULIN PUMP WITH CLOUD-BASED MONITORING AND EMERGENCY ALERT SYSTEM
60. REAL-TIME AI-DRIVEN HYPERTENSION RISK MONITORING WITH ALERT NOTIFICATION
61. SLEEP APNEA DETECTION SYSTEM USING SMART MATTRESS SENSORS AND EMBEDDED AI
62. IOT-ENABLED DIABETES TRACKING AND ALERT SYSTEM WITH MOBILE APP INTEGRATION
63. EDGE AI-BASED ALZHEIMER'S EARLY DETECTION SYSTEM THROUGH BEHAVIOR ANALYSIS
64. SMART RESPIRATORY HEALTH MONITORING WITH ENVIRONMENTAL PARAMETER ANALYSIS
65. ARTIFICIAL INTELLIGENCE-BASED ANEMIA DETECTION SYSTEM FROM BLOOD IMAGE DATA
66. EMBEDDED SMART SENSOR FOR NERVE GABA SIGNAL MONITORING AND ANALYSIS
67. AI-ENHANCED WOUND TYPE CLASSIFICATION AND SEVERITY GRADING SYSTEM
68. JOINT MOVEMENT TRACKER AND ARTHRITIS RISK DETECTION SYSTEM USING AI
69. BONE STRENGTH ANALYSIS DEVICE WITH AI-POWERED PREDICTIVE MODELING
70. BLINK-CONTROLLED WHEELCHAIR NAVIGATION FOR PARALYSIS-AFFECTED USERS
71. VOICE-ACTIVATED HOSPITAL BED ADJUSTMENT AND EMERGENCY CALL SYSTEM
72. AUTISM SPECTRUM DISORDER MONITORING USING DEEP LEARNING ON EEG AND AUDIO
73. WEARABLE PRESCRIPTION-CONTROLLED SMART TABLET DISPENSER FOR ELDERLY

74. SMART PRENATAL HEALTH MONITORING SYSTEM FOR HIGH-RISK PREGNANCIES
75. AI-BASED PERSON FINDER USING FACIAL RECOGNITION IN CCTV FOOTAGE
76. AI-POWERED HOME AUTOMATION SYSTEM WITH GESTURE AND VOICE CONTROL
77. COMA PATIENT BRAIN ACTIVITY MONITOR AND ALERT SYSTEM USING EEG
78. EYE BALL MOVEMENT-CONTROLLED VEHICLE FOR PHYSICALLY DISABLED DRIVERS
79. SMART PIEZOELECTRIC FLOOR MAT FOR INTRUDER DETECTION AND ALERT
80. HYDRATION LEVEL MONITORING DEVICE FOR ATHLETES USING SKIN CONDUCTANCE
81. VERTICAL WINDMILL-BASED POWER GENERATION AND REAL-TIME MONITORING SYSTEM
82. PRECISION AGRICULTURE MONITORING SYSTEM USING AI AND IMAGE PROCESSING
83. REHABILITATION GAME SYSTEM FOR STROKE PATIENTS USING HAND MOTION SENSORS
84. AI-POWERED PATIENT STETHOSCOPE SOUND CLASSIFICATION AND ANALYSIS SYSTEM
85. POSTURE MONITORING WEARABLE SYSTEM FOR SPINAL HEALTH IN DESK WORKERS
86. AI-BASED UNDERGROUND CABLE FAULT LOCATION AND TYPE DETECTION SYSTEM
87. PARKINSON'S GAIT ABNORMALITY DETECTION USING SMART INSOLES AND ML
88. AI-ENABLED LANE TRACING AND CRUISE CONTROL SYSTEM FOR SMART VEHICLES

89. YOGA MAT WITH PRESSURE AND POSTURE SENSORS FOR PERFORMANCE EVALUATION
90. FOREST ENVIRONMENT HEALTH MONITORING USING AI AND SATELLITE INTEGRATION
91. SMART BATTERY HEALTH MONITOR FOR BIOMEDICAL DEVICES
92. ELECTRIC HEATED HEALTH BAG WITH AI-CONTROLLED TEMPERATURE SETTINGS
93. SELF-NAVIGATING AI-BASED VACUUM CLEANER FOR MEDICAL FACILITIES
94. SWIMMER TRACKING AND EMERGENCY AIRBAG DEPLOYMENT SYSTEM FOR SAFETY
95. WIND ENERGY MONITORING SYSTEM USING IOT AND SMART METERS
96. SMART INFANT MONITORING SYSTEM WITH SIDS AND CRYING PATTERN DETECTION
97. SMART GYM ASSISTANT WITH AI-POWERED FORM CORRECTION AND REP COUNTER
98. AI-POWERED CATTLE HEALTH AND ACTIVITY MONITORING SYSTEM
99. ENDOCRINE SYSTEM FUNCTIONALITY MONITORING USING NEURAL NETWORKS
100. CELL BALANCING SYSTEM FOR MEDICAL POWER PACKS USING MACHINE LEARNING
101. AI-ENABLED SMART TDS AND WATER QUALITY MONITORING DEVICE
102. FACIAL SHAPE AND BMI BASED HEALTH MONITORING AND WELLNESS TRACKER
103. AI-BASED SCOLIOSIS DETECTION AND SPINE CURVE ANALYSIS FROM X-RAY IMAGES
104. BRAIN TUMOR DETECTION USING DEEP NEURAL NETWORKS ON MRI IMAGES
105. DIABETES RETINOPATHY DETECTOR USING CONVOLUTIONAL NEURAL NETWORKS

106. SMART E-HEALTH CARD WITH BLOCKCHAIN FOR TAMPER-PROOF MEDICAL RECORDS
107. IOT BASED NOISE POLLUTION MONITORING IN HOSPITALS FOR PATIENT COMFORT
108. AUTOMATED WHEELCHAIR NAVIGATION SYSTEM USING OBSTACLE DETECTION AI
109. REAL-TIME FATIGUE MONITORING SYSTEM FOR LONG-DISTANCE DRIVERS
110. EEG CONTROLLED APPLIANCE SYSTEM FOR QUADRIPLLEGIC PATIENTS
111. GAIT PATTERN LEARNING SYSTEM FOR AMPUTEE REHABILITATION FEEDBACK
112. POST-OPERATIVE RECOVERY TRACKING SYSTEM USING WEARABLE SENSORS
113. SMART MIRROR FOR VITAL MONITORING AND PERSONALIZED HEALTH TIPS
114. GESTURE-BASED COMMUNICATION AID FOR NON-VERBAL PATIENTS
115. BIOMETRIC ACCESS CONTROL WITH HEALTH PARAMETER VERIFICATION
116. NUTRIENT DEFICIENCY DETECTION USING FACIAL AND NAIL IMAGE ANALYSIS
117. UV RADIATION MONITORING WEARABLE FOR SKIN CANCER RISK ALERTS
118. DIGITAL STETHOSCOPE WITH AI FOR CARDIAC SOUND CLASSIFICATION
119. AI-BASED LIVER DISEASE DETECTION USING BLOOD CHEMISTRY PARAMETERS
120. NON-INVASIVE SMART GLUCOMETER USING NIR SPECTROSCOPY AND AI
121. EMOTION RECOGNITION SYSTEM FROM VOICE AND FACIAL MOVEMENTS FOR AUTISM

- 122. PULSE WAVE VELOCITY MEASUREMENT SYSTEM FOR VASCULAR HEALTH
- 123. AI-ENHANCED FALL DETECTION SYSTEM WITH MULTI-SENSOR FUSION
- 124. PATIENT EMOTION DETECTION USING REAL-TIME FACIAL EXPRESSION ANALYSIS
- 125. SMART EMERGENCY BRACELET WITH GPS, SOS, AND HEALTH ALERT
- 126. LUNG CAPACITY MONITORING USING SMART SPIROMETRY AND AI ANALYTICS
- 127. WIRELESS SMART THERMOMETER FOR CONTINUOUS FEVER MONITORING
- 128. REAL-TIME UTI DETECTION SYSTEM USING AI ON URINE TEST STRIP IMAGES
- 129. IOT AND AI-BASED HEARING LOSS MONITORING AND ALERT SYSTEM
- 130. AI-BASED EYE MOVEMENT TRACKING SYSTEM FOR NEUROLOGICAL DIAGNOSIS
- 131. ELECTROLYTE IMBALANCE DETECTION SYSTEM USING SWEAT SENSOR DATA
- 132. MACHINE LEARNING-BASED HEART DISEASE PREDICTION USING EHR DATA
- 133. AI-ENABLED ICU MONITORING DASHBOARD FOR REMOTE ACCESS
- 134. SMART HEALTH MONITORING SLIPPERS FOR DIABETIC NEUROPATHY DETECTION
- 135. REMOTE COGNITIVE HEALTH MONITORING SYSTEM FOR ELDERLY USING AI
- 136. CONTACTLESS THERMOMETER WITH VOICE OUTPUT FOR VISUALLY IMPAIRED
- 137. IOT AND AI-DRIVEN PHYSIOTHERAPY FEEDBACK SYSTEM
- 138. EEG-BASED BRAIN FATIGUE MONITORING FOR ACADEMIC AND OFFICE USE

139. AI-POWERED SKIN CANCER DETECTION SYSTEM USING DERMOSCOPIC IMAGES

140. RESPIRATORY INFECTION DETECTION USING COUGH PATTERN ANALYSIS

141. BLOOD PRESSURE MONITORING WITH REAL-TIME ALERTS VIA IOT APP

142. EMERGENCY RESPONSE DRONE WITH HEALTH MONITORING KIT FOR REMOTE AREAS

